Differentiable functions – 2

Mean value theorems: how to produce a point ξ

Homework

1 сентября 2026 года

Theoretical minimum. Rolle, Lagrange and Cauchy mean value theorems; the auxiliary function trick and the standard multipliers $e^{\lambda x}$, $\frac{1}{x}$, $\varphi(x)$; Darboux' theorem; repeated application of Rolle to a function with many zeros; the mean value theorem for integrals and its use in producing a point.

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1. (IMC 2025) Let $f\colon (0,\infty)\to\mathbb{R}$ be continuously differentiable and let $b>a>0$ satisfy $f(a)=f(b)=k$. Prove that there is $\xi\in(a,b)$ with $f(\xi)-\xi f'(\xi)=k$.

2. (RMO 1999) Let $a<b$ and let $f,g\colon[a,b]\to\mathbb{R}$ be differentiable with increasing derivatives and $f'(a)>0$, $g'(a)>0$. Prove that there is $c\in[a,b]$ with $\frac{f(b)-f(a)}{b-a}\cdot\frac{g(b)-g(a)}{b-a}=f'(c)g'(c)$.

3. (OMOUS 2022) Let $f\colon[0,1]\to\mathbb{R}$ be continuous with $\int_0^1 f(x)\,dx=\frac{\pi}{4}$. Prove that there is $x_0\in(0,1)$ with $\frac{1}{1+x_0}<f(x_0)<\frac{1}{2x_0}$.

4. Let f be continuous on $[a,b]$ and differentiable on (a,b), where $b-a\geqslant\pi$. Prove that there is $x_0\in(a,b)$ with $f'(x_0)<1+f(x_0)^2$.

5. (IMC 2016) Let $f\colon[a,b]\to\mathbb{R}$ be continuous on $[a,b]$ and differentiable on (a,b). Suppose f has infinitely many zeros but there is no $x\in(a,b)$ with $f(x)=f'(x)=0$.

   a) Prove that $f(a)f(b)=0$.

   b) Give an example of such a function on $[0,1]$.

6. (IMC 2005) Prove that for a three times differentiable $f\colon\mathbb{R}\to\mathbb{R}$ there is $\xi\in(-1,1)$ with $\frac{f'''(\xi)}{6}=\frac{f(1)-f(-1)}{2}-f'(0)$.

7. Let $f \in C^1[a,b]$ with $f > 0$. Prove that $\frac{f(b)-f(a)}{b-a} \leqslant \max_{x\in[a,b]}\Big(f(x) + \frac{(f'(x))^2}{4f(x)}\Big)$.

8. Let f be continuous on $[a,b]$ and differentiable on (a,b), where $0 < a < b$. Prove that there are $c_1, c_2 \in (a,b)$ with $\frac{f'(c_1)}{a+b} = \frac{f'(c_2)}{2c_2}$.

9. (IMC 1995) Let f be twice continuously differentiable on $(0,+\infty)$ with $\lim_{x\to 0^+} f'(x) = -\infty$ and $\lim_{x\to 0^+} f''(x) = +\infty$. Show that $\lim_{x\to 0^+} \frac{f(x)}{f'(x)} = 0$.

10. (IMC 2006) Find all sequences $a_0, a_1, \ldots, a_n$ of reals with $n \geqslant 1$ and $a_n \neq 0$ for which the following holds: if $f\colon \mathbb{R} \to \mathbb{R}$ is n times differentiable and $x_0 < x_1 < \cdots < x_n$ satisfy $f(x_0) = f(x_1) = \cdots = f(x_n) = 0$, then there is $h \in (x_0, x_n)$ with $a_0 f(h) + a_1 f'(h) + \cdots + a_n f^{(n)}(h) = 0$.